

Processing vs. Analysis

Image Formation

object in → image out

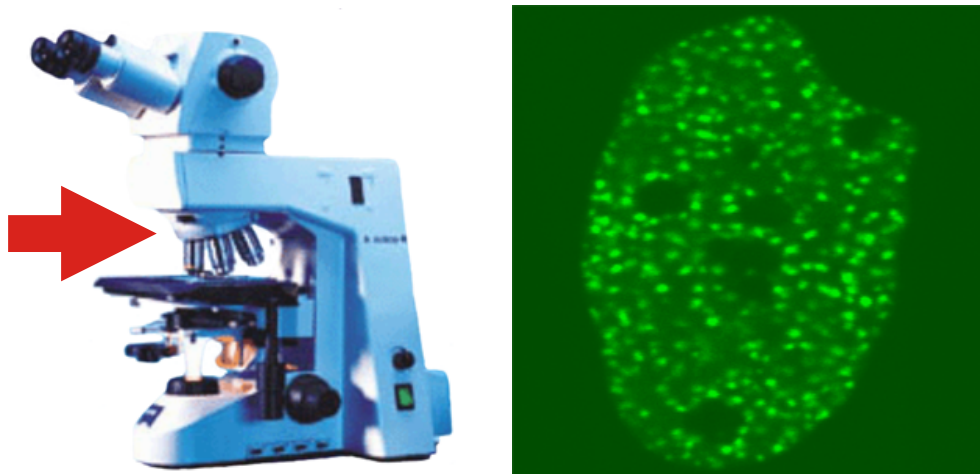


Image Processing

image in → image out

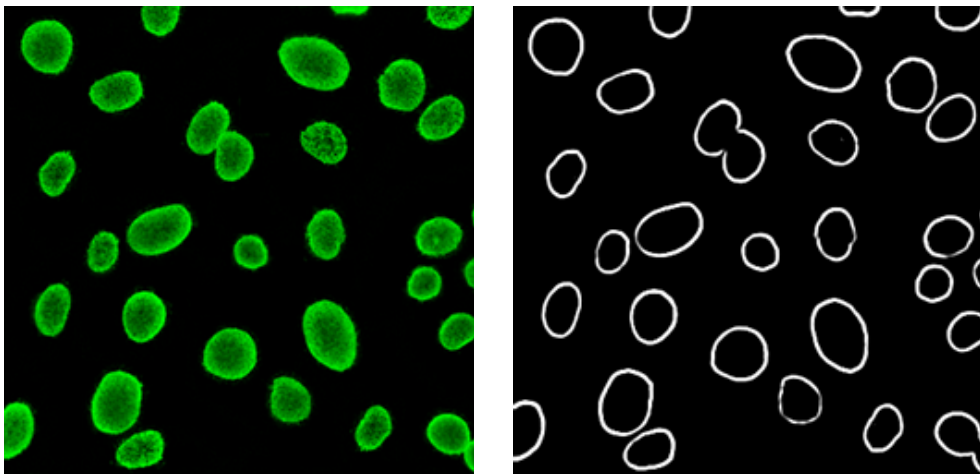
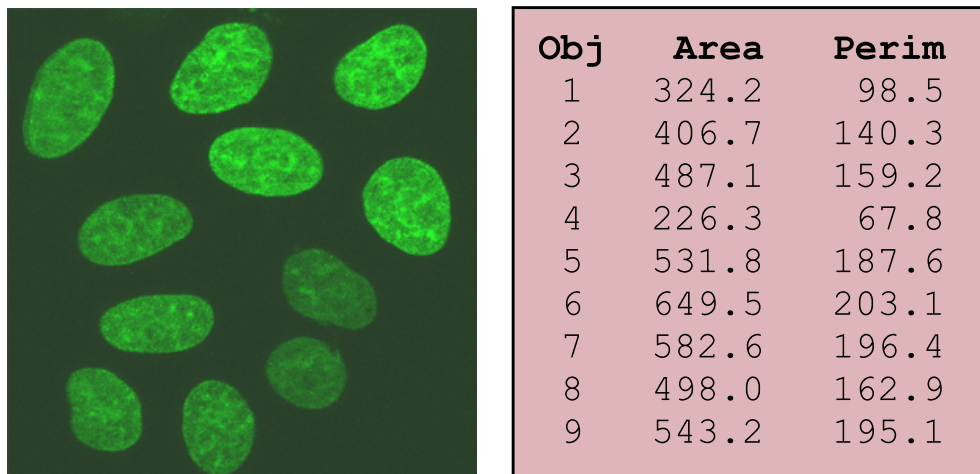


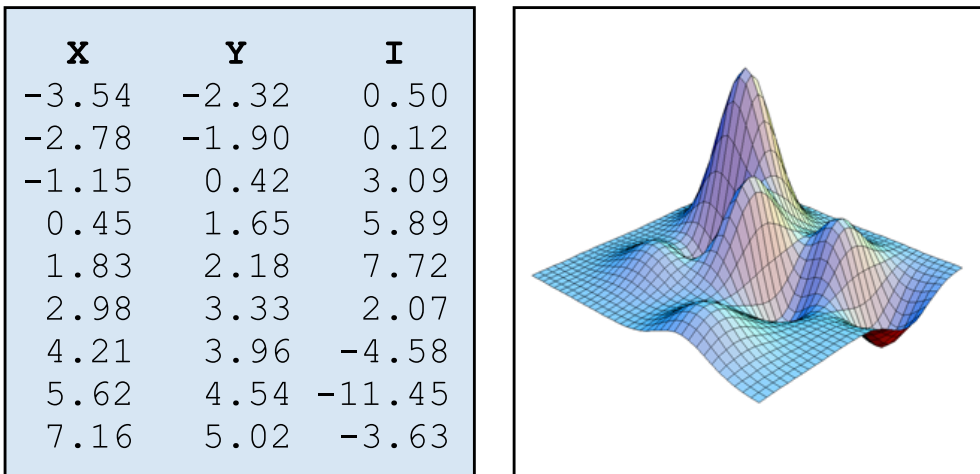
Image Analysis

image in → features out



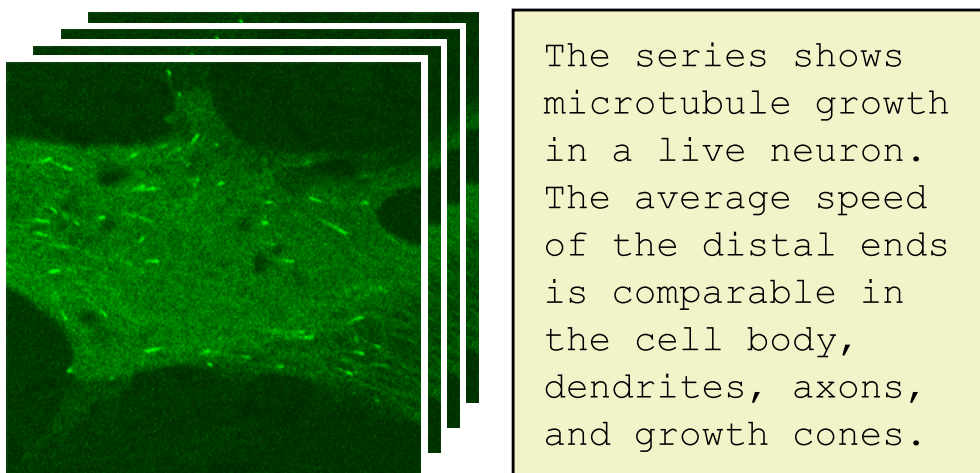
Computer Graphics

numbers in → image out



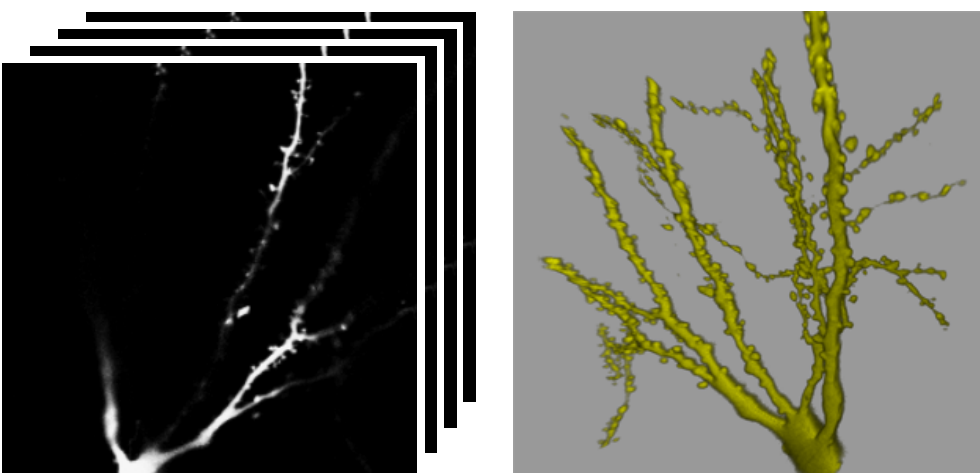
Computer Vision

image in → interpretation out



Visualization

image in → representation out



What we see...

...is *not* what we've got!

- Computer screens have 256 grey-levels (8-bit)
 - RGB: $256^3 = 16'777'216$ colours (24-bit, true-color)
 - This is about what the human eye can perceive
- Scientific images are often 16-bit = 65'536 grey-levels
 - The human eye can perceive less than that
- Your monitor shows ***less than 0.4%*** of what is in a good tiff-image